# Finite-Dimensional Vector Spaces

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## 1 *Auto-regressive model*

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Suppose that $z_1, z_2, \ldots$ is a time series, with the number $z_t$ giving the value in period or time $t$. For example $z_t$ could be the gross sales at a particular store on day $t$. An *auto-regressive* (AR) model is used to predict $z_{t+1}$ from the previous $M$ values, $z_t, z_{t-1}, \ldots, z_{t-M+1}$:

$$z_{t+1} = (z_t, z_{t-1}, \ldots z_{t-M+1})^{\mathrm{T}} \beta, \quad t = M, M+1, \ldots$$

Here $z_{t+1}$ denotes the AR model's prediction of $z_{t+1}$, $M$ is the memory length of the AR model, and the $M$-vector $\beta$ is the AR model coefficient vector. For this problem we will assume that the time period is daily, and $M = 10$. Thus, the AR model predicts tomorrow's value, given the values over the last 10 days.

For each of the following cases, give a short interpretation or description of the AR model in English, without referring to mathematical concepts like vectors, inner product, and so on. You can use words like 'yesterday' or 'today'.

(a) $\beta \approx e_1$.

(b) $\beta \approx 2e_1 - e_2$.

(c) $\beta \approx e_6$.

(d) $\beta \approx 0.5e_1 + 0.5e_2$.

### 1.1 Answer

**(a)** $\beta \approx e_1$

Tomorrow's value will be roughly the same as today's value. The model ignores everything that happened earlier in the past 10 days and simply predicts that whatever sales or numbers you had today will repeat tomorrow.

**(b)** $\beta \approx 2e_1 - e_2$

Tomorrow's value continues today's trend (the direction and amount of change from yesterday to today).

If sales grew by 100 between yesterday and today, the model predicts they will grow by another 100 tomorrow. Expressed as $e_1 + (e_1 - e_2)$ it takes today's value and adds today's change.

**(c)** $\beta \approx e_6$

Tomorrow's value will match the value from 6 days ago. The model looks back almost a full week and assumes tomorrow will look identical to whatever happened on that same day earlier in the cycle.

**(d)** $\beta \approx 0.5e_1 + 0.5e_2$

Tomorrow's value will be the simple average of today's and yesterday's values. The model smooths out day-to-day noise by taking a two-day moving average and using that to predict what comes next.